

Saransh Agrawal

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AI/ML Engineer with production experience in LLM infrastructure, agentic AI systems, and distributed training. Skilled in RAG pipelines, vector search, inference optimization, and ML platform development. Published at ACL, EACL. AWS Solutions Architect - Professional.

Education

Texas A&M University *Master of Science in Data Science: (GPA: 3.7/4.0)* *College Station, TX, May 2025*

Courses: Parallel Computing, Database Mgmt., Natural Language Processing, Reinforcement Learning, Data Stream & Algo.

Manipal Institute of Technology *Bachelors in Electronics and Instrumentation Engineering* *Aug 2021*

Courses: Data Structures & Algorithms, Microprocessors, Digital Image Processing, Embedded Systems, Digital System Design

Technical Skills

AI/ML Systems: PyTorch, DeepSpeed, vLLM, LangGraph, RAG Pipelines, Vector DBs (Qdrant)

Languages: Python, Rust, Go, JavaScript, C++, SQL

Infrastructure: AWS (SageMaker, Bedrock, EKS, Lambda, EC2), Kafka, Docker, Kubernetes, Terraform, HPC

Certificates: AWS Solutions Architect - Professional

Work Experience

AI Engineer Intern | Saptarishis Software (urpandit.com, astrologersdesk.net) **Nov 2025 – Present**

Early-stage startup building personalized conversational AI and recommendation systems *Remote, US*

- Built RAG pipeline combining vector retrieval (Qdrant), real-time API data, and user history for personalized conversational AI.
- Developing agentic matchmaking workflow with profile-based retrieval, conversational agents, and multi-step recommendation logic.
- Implemented PII guardrails using LLM safety techniques (constrained decoding, output filtering) to protect user privacy.

Research Assistant | FLAIR Lab (khhuang.me/group.html) **Aug 2024 – May 2025**

NLP Research Lab at Texas A&M University under Prof. Kuan-Hao Huang *College Station, TX*

- Built distributed LLM training pipeline (Llama3, Mistral-7B) on multi-node HPC with DeepSpeed, gaining 4x throughput.
- Reduced multimodal LLM (LLaVA) inference latency by 21% via KV-cache optimization (streaming LLM, cache merging).
- Architected automated LLM evaluation framework (MMLU, HELM) with distributed job scheduling across GPU configs.
- Led the Research on Unlearning of LLMs earning 2nd place in SemEval 2025 ACL Workshop.

Software Development Engineer | Viewzen Labs (viewzenlabs.com) **July 2021 – July 2023**

Enterprise MLaaS and data platform startup serving finance, healthcare, and government clients.

- Architected multi-tenant MLaaS platform with AutoML, experiment tracking, and Kafka-based inference pipelines for 50+ enterprise users.
- Built fault-tolerant ETL microservice (FastAPI, Kafka) with decoupled E/T/L stages, concurrent pipeline handling, and C++ transformations for throughput optimization.
- Developed AutoML feature enabling one-click baseline model generation (preprocessing, feature engineering, XGBoost) for rapid client onboarding.
- Deployed XGBoost risk model for maternal healthcare early intervention program, achieving 0.85 F1-score in production.
- Established MLOps practices: experiment tracking (MLflow), model registry, and automated retraining pipelines.

Machine Learning Engineer Intern | Centre for Digital Financial Inclusion (cdfi.in) **Feb 2021 – June 2021**

Non-profit developing tech solutions for financial inclusion, public service delivery in rural India.

- Built end-to-end Speech-to-SQL pipeline powering voice-activated analytics dashboard with sub-4s query-to-chart latency.
- Achieved 90% valid SQL generation rate using fine-tuned T5 with schema-aware decoding and query validation.
- Fine-tuned DeepSpeech ASR on Indian accent corpus, reducing WER to 17% for production transcription.

Publications

- [1] Saransh Agrawal and Kuan-Hao Huang. Selective Amnesia – Constrained Unlearning for Large Language Models via Knowledge Isolation. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025) at ACL*.
- [2] Ren-Wei Liang, Chin-Ting Hsu, Chan-Hung Yu, Saransh Agrawal, Shih-Cheng Huang, Shang-Tse Chen, Kuan-Hao Huang, and Shao-Hua Sun. Adaptive Helpfulness-Harmlessness Alignment with Preference Vectors. In *European Chapter of the Association for Computational Linguistics (EACL) 2026*